ORIGINAL ARTICLE

Trastuzumab Deruxtecan or Ramucirumab plus Paclitaxel in Gastric Cancer

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ABSTRACT

BACKGROUND

On the basis of phase 2 studies, trastuzumab deruxtecan was approved for patients with human epidermal growth factor receptor 2 (HER2)–positive metastatic gastric cancer or gastroesophageal junction adenocarcinoma who had previously received trastuzumab-based therapy. Ramucirumab plus paclitaxel is also a standard second-line treatment option regardless of HER2 status.

METHODS

We conducted an international, randomized, phase 3 trial comparing second-line trastuzumab deruxtecan at a dose of 6.4 mg per kilogram of body weight with ramucirumab plus paclitaxel in patients with HER2-positive metastatic gastric cancer or gastroesophageal junction adenocarcinoma confirmed on tumor biopsy conducted after the patient had progression while receiving trastuzumab-based therapy. The primary end point was overall survival. Secondary end points included progression-free survival, confirmed objective response (complete or partial response lasting ≥4 weeks), disease control, duration of response, and safety.

RESULTS

Among 494 patients who had undergone randomization, overall survival was significantly longer with trastuzumab deruxtecan than with ramucirumab plus paclitaxel (median, 14.7 vs. 11.4 months; hazard ratio for death, 0.70; 95% confidence interval, 0.55 to 0.90; P=0.004). Significant results were also seen with regard to progression-free survival (hazard ratio for disease progression or death, 0.74; 95% CI, 0.59 to 0.92) and confirmed objective response (in 44.3% of the patients in the trastuzumab deruxtecan group vs. 29.1% of those in the ramucirumab–paclitaxel group). The incidence of drug-related adverse events of any grade was 93.0% with trastuzumab deruxtecan and 91.4% with ramucirumab plus paclitaxel; the incidence of drug-related adverse events of grade 3 or higher was 50.0% and 54.1%, respectively. Adjudicated drug-related interstitial lung disease or pneumonitis occurred in 13.9% of the patients who received trastuzumab deruxtecan (grade 1 or 2 in 33 patients and grade 3 in 1) and in 1.3% of those who received ramucirumab plus paclitaxel (grade 3 in 2 patients and grade 5 in 1).

CONCLUSIONS

Trastuzumab deruxtecan led to significantly longer overall survival than ramucirumab plus paclitaxel among patients with HER2-positive metastatic gastric cancer or gastroesophageal junction adenocarcinoma. Adverse events were common in both groups. Events of interstitial lung disease or pneumonitis with trastuzumab deruxtecan, a known risk, were mainly low-grade. (Funded by Daiichi Sankyo and AstraZeneca; DESTINY-Gastric04 ClinicalTrials.gov number, NCT04704934.)

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*A list of the investigators in the DESTINY-Gastric04 trial is provided in the Supplementary Appendix, available at NEJM.org.

This article was published on May 31, 2025, at NEJM.org.

DOI: 10.1056/NEJMoa2503119

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Gastric cancer ranks fifth among all types of cancer in terms of prevalence and mortality.[1] An estimated 5 to 17% of gastric cancer cases are positive for human epidermal growth factor receptor 2 (HER2), defined as an immunohistochemical (IHC) score of 3+ (positive) or as an IHC score of 2+ (borderline) with in situ hybridization (ISH)–positive status.[2-5] For patients with unresectable or metastatic HER2-positive gastric cancer or gastroesophageal junction adenocarcinoma, the standard first-line therapy is chemotherapy plus trastuzumab.[6-8] On the basis of results of the recent KEYNOTE-811 trial, pembrolizumab can be added if programmed death ligand 1 is expressed (combined positive score, ≥1), in countries where this drug is available.[9,10]

Trastuzumab deruxtecan (also known as T-DXd or DS-8201) is an antibody–drug conjugate consisting of a humanized anti-HER2 monoclonal antibody linked to a topoisomerase I inhibitor payload by means of a tetrapeptide-based cleavable linker.[11,12] The randomized, phase 2 DESTINY-Gastric01 trial, which was conducted in Japan and South Korea, showed better response and longer overall survival with trastuzumab deruxtecan than with irinotecan or paclitaxel in the context of third-line or later therapy among patients with HER2-positive gastric cancer.[13] The single-group, phase 2 DESTINY-Gastric02 study (conducted in the United States and in European countries) and the DESTINY-Gastric06 study (conducted in China) also confirmed activity of trastuzumab deruxtecan in the context of second- and third-line therapy.[14,15] The results of these phase 2 trials led to the approval of trastuzumab deruxtecan in multiple countries for the treatment of patients with unresectable or metastatic HER2-positive gastric cancer or gastroesophageal junction adenocarcinoma who had previously received trastuzumab-based therapy.[6,7,16]

A confirmatory phase 3 trial of trastuzumab deruxtecan as second-line therapy is warranted to allow broader access of this treatment to patients around the world. Another standard second-line treatment option for patients with advanced or metastatic gastric cancer, regardless of HER2 status, is the combination of ramucirumab (an anti–vascular endothelial growth factor receptor 2 antibody) and paclitaxel.[6,7,17,18] Here, we describe the results of the primary efficacy analysis of the DESTINY-Gastric04 trial, a randomized, phase 3 trial designed to evaluate the efficacy and safety of trastuzumab deruxtecan as compared with ramucirumab plus paclitaxel as second-line therapy in patients with HER2-positive metastatic gastric cancer or gastroesophageal junction adenocarcinoma.

METHODS

TRIAL DESIGN AND TREATMENT

In this international, open-label, randomized, phase 3 trial, we enrolled eligible patients who had locally or centrally confirmed HER2-positive unresectable or metastatic gastric cancer or gastroesophageal junction adenocarcinoma on a tumor biopsy conducted after progression during trastuzumab-based therapy. We used this design in order to exclude patients who had loss of HER2 after trastuzumab-based therapy. Patients had to have an Eastern Cooperative Oncology Group performance-status score of 0 or 1 (on a scale from 0 to 5, with higher scores indicating greater disability). Patients were excluded if they had a history of noninfectious interstitial lung disease or pneumonitis for which treatment with glucocorticoids had been indicated, had interstitial lung disease or pneumonitis, or were suspected of having interstitial lung disease or pneumonitis that could not be ruled out by imaging (computed tomography) at the time of screening. The complete inclusion and exclusion criteria are listed in the protocol and in the Supplementary Appendix, both of which are available with the full text of this article at NEJM.org.

Patients were randomly assigned in a 1:1 ratio to receive intravenous infusions of either trastuzumab deruxtecan (6.4 mg per kilogram of body weight on day 1 of each 21-day cycle) or ramucirumab (8 mg per kilogram on days 1 and 15 of each 28-day cycle) plus paclitaxel (80 mg per square meter of body-surface area on days 1, 8, and 15 of each 28-day cycle). Randomization was stratified according to HER2 status after trastuzumab therapy (IHC 3+ vs. IHC 2+ and ISH-positive), geographic region (Asia [excluding mainland China] vs. western Europe vs. mainland China and the rest of the world), and time to progression during first-line therapy (<6 months vs. ≥6 months). Treatment continued until disease progression, withdrawal of patient consent, physician decision,

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or the occurrence of unacceptable adverse events. Crossover to the other treatment group was not planned during the trial period. However, after the trial, patients could receive treatment of the physician's choice, including trastuzumab deruxtecan in countries where it was available.

TRIAL OVERSIGHT

The trial was designed and sponsored by Daiichi Sankyo in collaboration with AstraZeneca and was approved by the institutional review board at each trial site. The trial was conducted in adherence with the principles of the Declaration of Helsinki, Good Clinical Practice guidelines of the International Council for Harmonisation, and local regulations on the conduct of clinical research. An external independent data and safety monitoring committee was established for this trial. The independent data and safety monitoring committee was responsible for reviewing efficacy data at the time of the interim analysis of overall survival as well as safety data at regular intervals during the conduct of the trial. All the patients provided written informed consent.

Data were analyzed and interpreted by the sponsor (Daiichi Sankyo) and the authors. All the authors reviewed the manuscript. The authors vouch for the analysis and accuracy of the data and for the fidelity of the trial to the protocol. Editorial and medical writing assistance was funded by the sponsor. AstraZeneca was not involved in data collection but was involved in data interpretation, as well as in the review of the manuscript and the approval of its submission for publication.

END POINTS

The primary end point was overall survival. Secondary end points included progression-free survival, confirmed objective response (complete or partial response lasting ≥4 weeks), disease control (confirmed objective response or stable disease for ≥6 weeks), and duration of response as determined by the investigator on the basis of the Response Evaluation Criteria in Solid Tumors, version 1.1. Exploratory end points included health-related quality of life with the use of patient-reported outcomes. Descriptions of the primary, secondary, and exploratory end points and the subgroups analyzed are provided in the Supplementary Appendix.

SAFETY

Adverse events were graded with the use of the Common Terminology Criteria for Adverse Events, version 5.0, of the National Cancer Institute and coded according to the *Medical Dictionary for Regulatory Activities*, version 27.1. The relatedness of adverse events to the trial drugs was determined by the investigator.

Interstitial lung disease or pneumonitis and left ventricular dysfunction were considered to be adverse events of special interest. Potential cases of interstitial lung disease or pneumonitis were evaluated by an independent adjudication committee. Confirmed cases were treated according to protocol-specified guidelines.

STATISTICAL ANALYSIS

The efficacy analysis was performed in the full analysis population, which included all the patients who had undergone randomization. The safety analysis population included all the patients who had received at least one dose of a trial drug. The primary efficacy analysis was the comparison of overall survival between the two trial groups on the basis of a stratified log-rank test with the use of the randomization stratification factors at an overall two-sided significance level of 0.05. We calculated that 339 deaths would provide the trial with 90% power to detect a hazard ratio of 0.70 and planned for a sample size of 490. A prespecified interim analysis was planned to occur after the completion of enrollment and the documentation of approximately 237 deaths (70% of the number of deaths specified for the final analysis). The final analysis was planned to take place after approximately 339 deaths only if no significant difference in overall survival was observed between the two groups at the interim analysis. The Lan–DeMets alpha-spending function with an O'Brien–Fleming boundary was applied to control the overall two-sided alpha at 0.05, and boundaries for significance were calculated on the basis of the number of observed events.

To control for multiplicity owing to the testing of multiple end points, the primary end point of overall survival and the secondary end points of progression-free survival and confirmed objective response were tested hierarchically with the use of a gatekeeping procedure. The hierarchy for this testing was overall survival, followed

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by progression-free survival, followed by confirmed objective response. Responses to patient-reported outcome questionnaires were analyzed descriptively. Details of additional statistical analyses are provided in the Supplementary Appendix.

RESULTS

PATIENTS

From May 21, 2021, through October 7, 2024, a total of 1088 patients at 152 sites underwent tissue screening for HER2 IHC expression, and 638

Table 1. Demographic and Clinical Characteristics of the Patients at Baseline.*

Characteristic	Trastuzumab Deruxtecan (N=246)	Ramucirumab + Paclitaxel (N=248)
Age		
Median (range) — yr†	63.2 (21.1–84.1)	64.3 (31.9–87.0)
Distribution — no. (%)		
<65 yr	140 (56.9)	130 (52.4)
≥65 yr	106 (43.1)	118 (47.6)
Male sex — no. (%)	187 (76.0)	205 (82.7)
Race — no. (%)‡		
White	116 (47.2)	130 (52.4)
Black	0	2 (0.8)
Asian	101 (41.1)	97 (39.1)
Other	28 (11.4)	19 (7.7)
Multiple	1 (0.4)	0
Primary tumor location — no. (%)		
Stomach	153 (62.2)	149 (60.1)
Gastroesophageal junction	93 (37.8)	99 (39.9)
ECOG performance-status score — no. (%)§		
0	97 (39.4)	88 (35.5)
1	148 (60.2)	158 (63.7)
2	1 (0.4)	1 (0.4)
Missing data	0	1 (0.4)
Geographic region, as a stratification factor — no. (%)¶		
Asia, excluding mainland China	57 (23.2)	60 (24.2)
Western Europe	140 (56.9)	139 (56.0)
Mainland China or rest of world	49 (19.9)	49 (19.8)
Region — no. (%)‖		
Asia, including mainland China	97 (39.4)	95 (38.3)
Europe	123 (50.0)	140 (56.5)
Rest of world	26 (10.6)	13 (5.2)
HER2 status — no. (%)**		
IHC 2+ and ISH-positive	39 (15.9)	40 (16.1)
IHC 3+	207 (84.1)	208 (83.9)
Time to progression during first-line therapy — no. (%)		
<6 mo	61 (24.8)	61 (24.6)
≥6 mo	185 (75.2)	187 (75.4)

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Table 1. (Continued.)

Characteristic	Trastuzumab Deruxtecan (N=246)	Ramucirumab + Paclitaxel (N=248)
No. of metastatic sites — no. (%)		
<2	73 (29.7)	75 (30.2)
≥2	173 (70.3)	173 (69.8)
Liver metastases — no. (%)	147 (59.8)	158 (63.7)
Brain metastases — no. (%)	16 (6.5)	18 (7.3)
Previous treatment with an immune checkpoint inhibitor — no. (%)	39 (15.9)	38 (15.3)

* Geographic region, human epidermal growth factor receptor 2 (HER2) status, and time to progression during first-line therapy were the stratification factors used by the interactive response technology. Percentages may not total 100 because of rounding.

† The following formula was used to calculate patient age in years: (date of informed consent − date of birth + 1) ÷ 365.25.

‡ Race was reported by the patient.

§ Eastern Cooperative Oncology Group (ECOG) performance-status scores range from 0 to 5, with higher scores indicating greater disability. A score of 0 indicates that the patient is fully active, a score of 1 that the patient is restricted in physically strenuous activity, and a score of 2 that the patient is ambulatory and capable of all self-care.

¶ The countries included in the categories of Asia (except mainland China), western Europe, and mainland China and the rest of world according to geographic region (stratification factor) were as follows: Asia included Hong Kong, Japan, Singapore, South Korea, and Taiwan; western Europe included Belgium, France, Germany, Ireland, Israel, Italy, Portugal, Spain, Turkey, and the United Kingdom; and mainland China and rest of the world included Argentina, Brazil, Chile, China, Poland, and Romania.

‖ The countries included in the categories of Europe and the rest of world according to region were as follows: Europe included Belgium, France, Germany, Ireland, Italy, Poland, Portugal, Romania, Spain, and the United Kingdom; and the rest of world included Argentina, Brazil, Chile, Israel, and Turkey.

** HER2 status was defined as a HER2 immunohistochemical (IHC) analysis of a score of 3+ (positive) or a score of 2+ (borderline) and positive results on in situ hybridization (ISH).

patients underwent main screening for additional eligibility criteria (Fig. S1 in the Supplementary Appendix). A total of 494 patients were randomly assigned to receive trastuzumab deruxtecan (246 patients) or ramucirumab plus paclitaxel (248 patients). The demographic and baseline clinical characteristics of the patients were balanced between the two groups (Table 1). The representativeness of the trial participants is shown in Table S1. Black patients were underrepresented in this trial. As of the data-cutoff date for the interim analysis (October 24, 2024), a total of 18.9% of the patients in the trastuzumab deruxtecan group and 18.5% of those in the ramucirumab–paclitaxel group were continuing to receive treatment.

EFFICACY

As of the data-cutoff date, 124 patients (50.4%) in the trastuzumab deruxtecan group and 142 patients (57.3%) in the ramucirumab–paclitaxel group had died; the median duration of follow-up for the analysis of overall survival was 16.8 months (95% confidence interval [CI], 14.0 to 20.0) and 14.4 months (95% CI, 13.1 to 19.7), respectively. Overall survival was significantly longer in the trastuzumab deruxtecan group than in the ramucirumab–paclitaxel group (median, 14.7 months [95% CI, 12.1 to 16.6] vs. 11.4 months [95% CI, 9.9 to 15.5]; hazard ratio for death, 0.70; 95% CI, 0.55 to 0.90; P=0.004, which crossed the prespecified O'Brien–Fleming boundary [<0.0228 on the basis of the number of deaths]) (Fig. 1A and Table 2). The estimated overall survival was 83.5% in the trastuzumab deruxtecan group and 74.4% in the ramucirumab–paclitaxel group at 6 months and 57.6% and 48.9%, respectively, at 12 months. The estimated overall survival at 24 months was 29.0% in the trastuzumab deruxtecan group and 13.9% in the ramucirumab–paclitaxel group. Additional survival analyses are shown in Figure S2 and Table S2.

Progression-free survival was also significantly longer with trastuzumab deruxtecan than with

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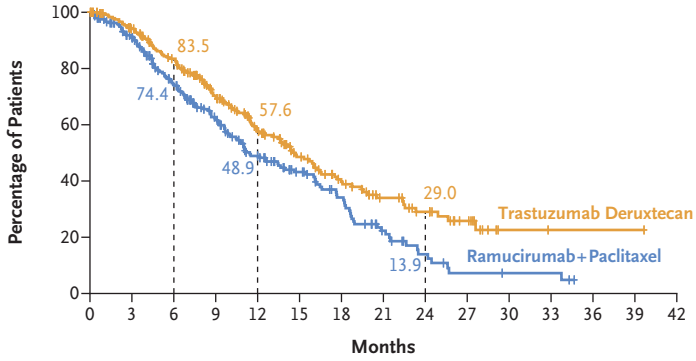
A Overall Survival



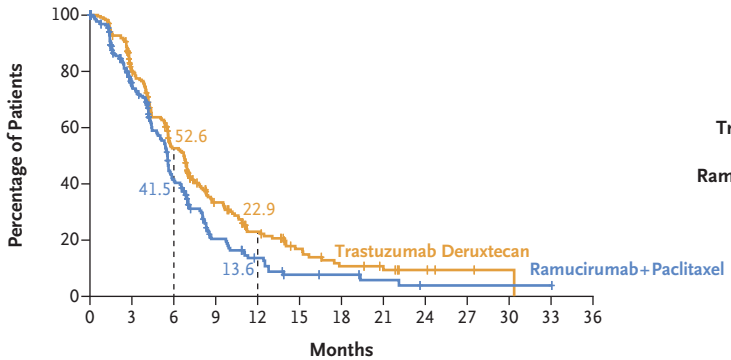
	No. of Deaths	Median Overall Survival (95% CI)
		mo
Trastuzumab Deruxtecan	124	14.7 (12.1–16.6)
Ramucirumab+ Paclitaxel	142	11.4 (9.9–15.5)

Hazard ratio for death, 0.70 (95% CI, 0.55–0.90)
P=0.004

No. at Risk

Trastuzumab deruxtecan	246	219	185	134	94	65	45	30	21	12	2	1	1	1	0
Ramucirumab+ paclitaxel	248	204	150	109	76	52	36	18	9	4	3	3	0		

B Progression-free Survival



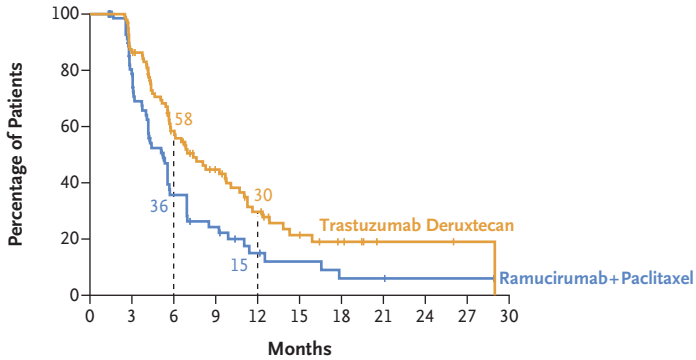
	No. of Events	Median Progression-free Survival (95% CI)
		mo
Trastuzumab Deruxtecan	166	6.7 (5.6–7.1)
Ramucirumab+ Paclitaxel	156	5.6 (4.9–5.8)

Hazard ratio for disease progression or death, 0.74 (95% CI, 0.59–0.92)
P=0.007

No. at Risk

Trastuzumab deruxtecan	246	173	102	51	30	17	10	7	4	2	1	0
Ramucirumab+ paclitaxel	248	144	68	25	14	6	5	3	1	1	1	1	0

C Duration of Response



	No. of Events	Median Duration of Response (95% CI)
		mo
Trastuzumab Deruxtecan	62	7.4 (5.7–10.1)
Ramucirumab+ Paclitaxel	52	5.3 (4.1–5.7)

No. at Risk

Trastuzumab deruxtecan	104	81	45	29	17	10	6	2	2	1	0
Ramucirumab+ paclitaxel	69	50	19	12	6	4	2	2	1	1	0

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Figure 1 (facing page). Overall Survival, Progression-free Survival, and Duration of Response.

In the analysis of overall survival (Panel A), 124 patients (50.4%) in the trastuzumab deruxtecan group and 142 patients (57.3%) in the ramucirumab–paclitaxel group died. Other patients had their data censored (tick marks). The two-sided P value of 0.004 crossed the O'Brien–Fleming boundary of significance (<0.0228 on the basis of the number of deaths). In the analysis of progression-free survival as assessed by the investigator (Panel B), 149 patients (60.6%) in the trastuzumab deruxtecan group and 127 patients (51.2%) in the ramucirumab–paclitaxel group had progressive disease, and 17 patients (6.9%) and 29 patients (11.7%), respectively, died. Other patients had their data censored (tick marks). In the analysis of duration of response (Panel C), 59 of 104 patients with a response (57%) in the trastuzumab deruxtecan group and 46 of 69 (67%) in the ramucirumab–paclitaxel group had progressive disease, and 3 patients (3%) and 6 patients (9%), respectively, died. Landmark analyses at 6 months and 12 months are shown in all panels. A landmark analysis at 24 months is also shown for overall survival (Panel A).

ramucirumab plus paclitaxel (median, 6.7 months [95% CI, 5.6 to 7.1] vs. 5.6 months [95% CI, 4.9 to 5.8]; hazard ratio for disease progression or death, 0.74; 95% CI, 0.59 to 0.92; P=0.007, which crossed the prespecified boundary [<0.0185 on the basis of the number of events]) (Fig. 1B and Table 2). The estimated progression-free survival was 52.6% in the trastuzumab deruxtecan group and 41.5% in the ramucirumab–paclitaxel group at 6 months and was 22.9% and 13.6%, respectively, at 12 months.

A confirmed objective response occurred in 44.3% of the patients (95% CI, 37.8 to 50.9) in the trastuzumab deruxtecan group and in 29.1% of those (95% CI, 23.4 to 35.3) in the ramucirumab–paclitaxel group (difference, 15.1 percentage points; P<0.001) (Table 2). The median duration of response was 7.4 months (95% CI, 5.7 to 10.1) with trastuzumab deruxtecan and 5.3 months (95% CI, 4.1 to 5.7) with ramucirumab plus paclitaxel (Fig. 1C). Response was maintained in 58% of the patients in the trastuzumab deruxtecan group and in 36% of those in the ramucirumab–paclitaxel group at 6 months and in 30% and 15%, respectively, at 12 months.

The number of patients in each subgroup was small, which precluded conclusions about the generalizability of the results (Fig. S3). However, 13 of the 23 subgroups in the analysis of overall survival and 16 of the 23 subgroups in the analysis of progression-free survival had upper boundaries of the confidence interval of 1.00 or more. More data are necessary to evaluate the suggested differences regarding efficacy in Asia (including mainland China) and with respect to the primary tumor location (stomach or gastroesophageal junction).

After the end of the trial treatment period, 126 patients (51.2%) in the trastuzumab deruxtecan group and 118 patients (47.6%) in the ramucirumab–paclitaxel group received subsequent anticancer treatment of the physician's choice. Three patients (1.2%) in the trastuzumab deruxtecan group were treated with commercially available trastuzumab deruxtecan after discontinuing trial treatment; 5 patients (2.0%) were treated with disitamab vedotin, an anti-HER2 antibody–drug conjugate approved in China. In the ramucirumab–paclitaxel group, 64 patients (25.8%) received trastuzumab deruxtecan or disitamab vedotin after discontinuing the trial treatment (Table S3). Additional analyses of overall survival, including a rank-preserving structural failure time model analysis to adjust the overall survival among patients in the ramucirumab–paclitaxel group who received trastuzumab deruxtecan or disitamab vedotin after the trial treatment was discontinued, are shown in Table S2. More patients in Asia than in other regions received subsequent HER2-directed antibody–drug conjugate therapies after the trial (Table S4).

SAFETY

The median duration of treatment was 5.4 months (range, 0.7 to 30.3) with trastuzumab deruxtecan and 4.6 months (range, 0.9 to 34.9) with ramucirumab plus paclitaxel. A total of 244 patients in the trastuzumab deruxtecan group and 233 in the ramucirumab–paclitaxel group were included in the safety analysis population. Drug-related adverse events were reported in 93.0% of the patients treated with trastuzumab deruxtecan and in 91.4% of those treated with ramucirumab plus paclitaxel (Table S5). Drug-related serious adverse events occurred in 45 patients (18.4%) who received trastuzumab deruxtecan and in 41 patients (17.6%) who received ramucirumab plus

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Table 2. Efficacy Summary.

End Point	Trastuzumab Deruxtecan (N=246)	Ramucirumab + Paclitaxel (N=248)	Treatment Effect (95% CI)*	P Value†
Overall survival (95% CI)‡				
Median — mo	14.7 (12.1–16.6)	11.4 (9.9–15.5)	0.70 (0.55–0.90)	0.004
At 6 mo — %	83.5 (78.0–87.7)	74.4 (68.0–79.7)		
At 12 mo — %	57.6 (50.3–64.2)	48.9 (41.5–55.9)		
At 24 mo — %	29.0 (21.3–37.2)	13.9 (7.8–21.8)		
Progression-free survival (95% CI)§				
Median — mo	6.7 (5.6–7.1)	5.6 (4.9–5.8)	0.74 (0.59–0.92)	0.007
At 6 mo — %	52.6 (45.7–59.1)	41.5 (34.3–48.7)		
At 12 mo — %	22.9 (16.9–29.6)	13.6 (8.4–20.0)		
Response				
Confirmed objective response¶				
No. of patients with response/total no.	104/235	69/237		
Percent of patients with response (95% CI)	44.3 (37.8–50.9)	29.1 (23.4–35.3)	15.1 (6.1–24.2)	<0.001
Confirmed best overall response — no./total no. (%)				
Complete response‖	7/235 (3.0)	3/237 (1.3)		
Partial response	97/235 (41.3)	66/237 (27.8)		
Stable disease**	112/235 (47.7)	111/237 (46.8)		
Progressive disease	13/235 (5.5)	22/237 (9.3)		
Could not be evaluated	6/235 (2.6)	35/237 (14.8)		
Disease control¶				
No. of patients with disease control/total no.	216/235	180/237		
Percent of patients with disease control (95% CI)	91.9 (87.7–95.1)	75.9 (70.0–81.2)	16.0 (9.1–22.9)	
Duration of response (95% CI)††				
Median — mo	7.4 (5.7–10.1)	5.3 (4.1–5.7)		
At 6 mo — %	58 (47–68)	36 (24–48)		
At 12 mo — %	30 (19–41)	15 (7–26)		

* The treatment effect in the analysis of overall survival is a hazard ratio for death, in the analysis of progression-free survival is a hazard ratio for disease progression or death, and in the analyses of overall response and disease control is an absolute difference (in percentage points).

† Two-sided P values for the analyses of overall survival and progression-free survival were calculated by means of a stratified log-rank test, and the stratified Cox proportional-hazards model was adjusted for the HER2 status stratification factor (IHC 3+ vs. IHC 2+ and ISH-positive). The P value for the analysis of overall response was calculated with the use of the Cochran–Mantel–Haenszel test with adjustment for the HER2 status stratification factor.

‡ The median overall survival and overall survival at specified time points were calculated by a Kaplan–Meier analysis. The confidence interval for the median was calculated with the use of the Brookmeyer–Crowley method. The confidence intervals for overall survival at specified time points were calculated with the use of the Greenwood formulation.

§ The median progression-free survival was calculated by a Kaplan–Meier analysis, and the confidence interval for the median was calculated with the use of the Brookmeyer–Crowley method. Estimates and confidence intervals for progression-free survival at the specified time points were calculated by a Kaplan–Meier analysis.

¶ The patients who were eligible for the analysis of overall response were those who underwent randomization at least 77 days (i.e., 2×6 weeks−1 week) before the data-cutoff date for the interim analysis. The percentages of patients with a confirmed objective response (complete or partial response lasting ≥4 weeks) and disease control (confirmed objective response or stable disease for ≥6 weeks) were calculated with the use of the number of eligible patients as the denominator. The confidence intervals were calculated on the basis of the Clopper–Pearson method for single proportion. The two-sided 95% confidence interval for the difference in overall response was based on Wald method with the use of continuity correction.

‖ Patients with a complete response who did not have target lesions at baseline were included in this analysis.

** Patients without target lesions at baseline who did not have a complete response or did not have progressive disease were included in stable disease category.

†† The analysis of duration of response included only patients with a partial response or better. The median duration of response was calculated by a Kaplan–Meier analysis, and the confidence interval for the median was calculated with the use of the Brookmeyer–Crowley method. Estimates and confidence intervals for duration of response at the specified time points were calculated by a Kaplan–Meier analysis.

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paclitaxel; drug-related adverse events of grade 3 or higher occurred in 122 patients (50.0%) and 126 patients (54.1%), respectively. The most common drug-related adverse events of grade 3 or higher (occurring in ≥10% of the patients in either group) were neutropenia (in 28.7% of the patients) and anemia (in 13.9%) with trastuzumab deruxtecan and neutropenia (in 35.6%), anemia (in 13.7%), and leukopenia (in 12.4%) with ramucirumab plus paclitaxel (Table 3).

Drug-related adverse events were associated with drug discontinuation in 11.5% of the patients who received trastuzumab deruxtecan and in 13.3% of those who received ramucirumab plus paclitaxel; with dose reduction in 31.1% and 36.1%, respectively; and with dose interruption in 38.5% and 51.1%, respectively (Table S5). Drug-related adverse events that were associated with death occurred in four patients (1.6%) who received trastuzumab deruxtecan (upper gastrointestinal hemorrhage, intestinal obstruction, sudden death, and death not otherwise specified in one patient each) and in two patients (0.9%) who received ramucirumab plus paclitaxel (gastric perforation and interstitial lung disease in one patient each).

Drug-related interstitial lung disease or pneumonitis, as adjudicated by an independent committee, occurred in 34 patients (13.9%) who received trastuzumab deruxtecan (grade 1 in 7 patients [2.9%], grade 2 in 26 [10.7%], and grade 3 in 1 [0.4%]) and in 3 patients (1.3%) who received ramucirumab plus paclitaxel (grade 3 in 2 patients [0.9%] and grade 5 in 1 [0.4%]) (Table S6). Left ventricular dysfunction occurred in 6 patients (2.5%) who received trastuzumab deruxtecan (grade 2 and grade 3 events in 3 patients each) and in 4 patients (1.7%) who received ramucirumab plus paclitaxel (grade 1 and grade 2 events in 2 patients each).

PATIENT-REPORTED OUTCOMES

Health-related quality-of-life questionnaires that were completed by the patients reflected that quality of life was maintained during the course of treatment with both trastuzumab deruxtecan and ramucirumab plus paclitaxel, as assessed according to changes from baseline data on the EuroQol 5-Dimension 5-Level visual analogue scale and the Functional Assessment of Cancer Therapy–Gastric subscale. Scores were similar in the two groups (Fig. S4).

DISCUSSION

The phase 3 DESTINY-Gastric04 trial showed significantly prolonged overall survival with a HER2-directed therapy as second-line therapy in patients with HER2-positive unresectable or metastatic gastric cancer or gastroesophageal junction adenocarcinoma. The improved efficacy of trastuzumab deruxtecan over ramucirumab plus paclitaxel was further supported by significant improvements in secondary end points, including progression-free survival and response — findings similar to what has been reported previously with trastuzumab deruxtecan in phase 2 studies.[13-15]

In-study crossover between the treatment groups was not planned in this trial. However, patients did receive a range of subsequent anticancer systemic therapies, including anti-HER2 antibody–drug conjugates, such as trastuzumab deruxtecan and disitamab vedotin, given the availability of these drugs as approved third-line options in some countries. Nevertheless, a significant benefit in overall survival was observed with trastuzumab deruxtecan as compared with ramucirumab plus paclitaxel.

In the current trial, overall survival and progression-free survival with ramucirumab plus paclitaxel were longer in our trial than in the overall population of the RAINBOW trial (median overall survival, 11.4 vs. 9.6 months; median progression-free survival, 5.6 vs. 4.4 months).[17] However, cross-trial comparisons should be interpreted with caution, given that the differences may be due to variations in trial design (e.g., in the RAINBOW trial, patients were enrolled regardless of HER2 status) and the post-trial therapies used.[17] In the current trial, the median overall survival benefit exceeded the median progression-free survival benefit, a finding consistent with results in the DESTINY-Gastric01 trial.[13] The greater percentage of patients with a response and tumor shrinkage with trastuzumab deruxtecan than with ramucirumab plus paclitaxel may have contributed to longer overall survival (prolonged postprogression survival), as seen in other trials.[19,20] Thus, after the failure of other HER2-directed thera-

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Table 3. Most Common Drug-Related Adverse Events.*

Event	Trastuzumab Deruxtecan (N=244)		Ramucirumab + Paclitaxel (N=233)	
	Any Grade	Grade ≥3	Any Grade	Grade ≥3
	number of patients with event (percent)			
Any drug-related adverse event	227 (93.0)	122 (50.0)	213 (91.4)	126 (54.1)
Fatigue†	117 (48.0)	17 (7.0)	88 (37.8)	6 (2.6)
Neutropenia‡	117 (48.0)	70 (28.7)	114 (48.9)	83 (35.6)
Nausea	108 (44.3)	12 (4.9)	33 (14.2)	0
Anemia§	76 (31.1)	34 (13.9)	77 (33.0)	32 (13.7)
Decreased appetite	71 (29.1)	5 (2.0)	42 (18.0)	3 (1.3)
Leukopenia¶	65 (26.6)	18 (7.4)	52 (22.3)	29 (12.4)
Thrombocytopenia‖	65 (26.6)	21 (8.6)	32 (13.7)	7 (3.0)
Diarrhea	63 (25.8)	3 (1.2)	47 (20.2)	5 (2.1)
Alopecia	59 (24.2)	0	62 (26.6)	0
Aminotransferase level increased**	53 (21.7)	5 (2.0)	22 (9.4)	1 (0.4)
Vomiting	49 (20.1)	7 (2.9)	16 (6.9)	1 (0.4)
Interstitial lung disease or pneumonitis††	34 (13.9)	1 (0.4)	3 (1.3)	3 (1.3)
Weight decreased	27 (11.1)	3 (1.2)	9 (3.9)	1 (0.4)
Constipation	26 (10.7)	0	12 (5.2)	0
Lymphopenia‡‡	25 (10.2)	5 (2.0)	13 (5.6)	3 (1.3)
Stomatitis§§	12 (4.9)	1 (0.4)	33 (14.2)	1 (0.4)
Neuropathy¶¶	6 (2.5)	0	68 (29.2)	8 (3.4)
Epistaxis	4 (1.6)	0	30 (12.9)	1 (0.4)
Hypertension	1 (0.4)	0	36 (15.5)	19 (8.2)

* Shown are adverse events that emerged or worsened after the initiation of a trial drug until 47 days after the last dose of a trial drug. Adverse events that occurred in at least 10% of the patients with any grade of drug-related adverse event in either treatment group are listed according to preferred or grouped term. Adverse events were coded with the use of the *Medical Dictionary for Regulatory Activities*, version 27.1, and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0. If a patient had more than one event according to a preferred term or grouped term, the patient was counted once at each level of summation. The preferred terms and grouped terms are shown in a descending order of frequency in the trastuzumab deruxtecan group.

† The group term fatigue includes fatigue, asthenia, malaise, and lethargy.

‡ The group term neutropenia includes neutrophil count decreased and neutropenia.

§ The group term anemia includes hemoglobin decreased, red-cell count decreased, anemia, and hematocrit decreased.

¶ The group term leukopenia includes white-cell count decreased and leukopenia.

‖ The group term thrombocytopenia includes platelet count decreased and thrombocytopenia.

** The group term aminotransferase level increased includes aminotransferase levels increased, aspartate aminotransferase increased, alanine aminotransferase increased, γ-glutamyltransferase increased, liver-function test abnormal, hepatic function abnormal, and liver-function test increased.

†† Interstitial lung disease includes all events that were considered by the adjudication committee as being related to a trial drug.

‡‡ The group term lymphopenia includes lymphocyte count decreased and lymphopenia.

§§ The group term stomatitis includes stomatitis, aphthous ulcer, mouth ulceration, oral mucosa erosion, oral mucosal blistering, and oral mucosal eruption.

¶¶ The group term neuropathy includes peripheral sensory neuropathy, neuropathy peripheral, neurotoxicity, peripheral motor neuropathy, and polyneuropathy.

pies in the context of second-line treatment of metastatic gastric cancer or gastroesophageal junction adenocarcinoma,[21,22] the clinical benefit that was observed with trastuzumab deruxtecan over ramucirumab plus paclitaxel in our trial suggests that trastuzumab deruxtecan may be the preferred second-line treatment option in patients whose tumors retain HER2 expression

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after progression occurs during a trastuzumab-containing regimen.

A high percentage of patients in this trial had adverse events. In general, the incidence of adverse events was similar in the two groups, and no new safety signals were noted. However, trastuzumab deruxtecan has a risk of interstitial lung disease or pneumonitis that requires careful monitoring and that led to the exclusion of patients with a history of pulmonary problems.[11,13-15] The most common drug-related adverse events with trastuzumab deruxtecan were fatigue or were gastrointestinal or hematologic in nature, whereas ramucirumab plus paclitaxel was more commonly associated with hematologic adverse events, fatigue, and neuropathy. Prophylactic antiemetic measures for trastuzumab deruxtecan therapy that involved a combination of antiemetic agents, such as dexamethasone with a 5-hydroxytryptamine receptor antagonist or a neurokinin-1 receptor antagonist, were recommended according to the protocol (see the Supplementary Appendix), and the incidence of nausea of grade 3 or higher was low. The incidence of serious adverse events or drug-related adverse events leading to drug discontinuation was similar in the two trial groups.

Interstitial lung disease or pneumonitis is an important risk associated with trastuzumab deruxtecan therapy. The incidence of interstitial lung disease or pneumonitis with trastuzumab deruxtecan in this trial was similar to that seen in previous studies in gastric cancer with trastuzumab deruxtecan at a dose of 6.4 mg per kilogram or in other studies in other tumor types treated with a dose of 5.4 mg per kilogram; most events in this trial were of grade 1 or 2.[11,13-15,23-25] Patients with a history of interstitial lung disease or pneumonitis were excluded from this trial. To ensure early detection and effective treatment of interstitial lung disease or pneumonitis, published guidelines recommend proactive monitoring of signs and symptoms, the use of imaging to identify potential cases, prompt dose interruption, and early initiation of glucocorticoid treatment.[26]

This trial has several limitations. The different treatment schedules of the trial drugs necessitated an open-label design. In addition, the trial was not designed or powered to assess significance in subgroups, and some subgroups contained a relatively small number of patients. Finally, the tumor biopsy samples that were obtained may not reflect the HER2 status of all metastases, owing to intrapatient heterogeneity. However, given that the loss of HER2 expression after trastuzumab-containing regimens has been documented and the usefulness of rebiopsy was suggested,[27-29] this trial required rebiopsy after progression occurred during previous treatment with trastuzumab in order to confirm HER2 status. These results support the current recommendation and consideration of rebiopsy to confirm HER2 status before trastuzumab deruxtecan treatment, if feasible.[6,7,16,30]

This trial showed that trastuzumab deruxtecan had superior efficacy as second-line therapy over ramucirumab plus paclitaxel in patients with HER2-positive unresectable or metastatic gastric cancer or gastroesophageal junction adenocarcinoma. These results warrant further evaluation of trastuzumab deruxtecan in the context of first-line therapy.

Supported by Daiichi Sankyo and AstraZeneca.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

We thank the patients and their families for their participation; the trial-site staff for their contributions; Meredith Venerus, M.Sc., for assistance with the patient-reported outcomes data; and Elize Wolmarans, Ph.D., Vishal Gor, Ph.D., M.Pharm., C.M.P.P., Laura Halvorson, Ph.D., and Jill Shults, Ph.D., all of ApotheCom, for medical writing and editorial assistance, funded by Daiichi Sankyo, with an earlier version of the manuscript.

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